Run report: cen-q7-nonsplit-run-20261024T2243Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (sampled) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: cen-q7-nonsplit 20261024T2243Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 efce43f3fdaa848389a03f3d778a9b9604fe23d1d6e5e1bd75e777a49fb6cf01
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 26415a19d8d07de211c7992a6c97ca99a2a3a358fabb5c02e7893639121efd5b
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/cen-q7-nonsplit-run-20261024T2243Z CENSUS 7 nonsplit 30000000 202610247 1 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 522
```

tool_log.txt

```
nonsplit 7: 6825623/30000000 samples, res-shaped 216, allowed-trace cand 216, smooth hits 216, 120s
nonsplit 7: 13749106/30000000 samples, res-shaped 415, allowed-trace cand 415, smooth hits 415, 240s
nonsplit 7: 20641572/30000000 samples, res-shaped 591, allowed-trace cand 591, smooth hits 591, 360s
nonsplit 7: 27543164/30000000 samples, res-shaped 781, allowed-trace cand 781, smooth hits 781, 480s
nonsplit q=7: samples=30000000 (random) res_shaped=839 allowed_trace_candidates=839 smooth_prime_order_hits=839 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=0/0 secs=522.0
nonsplit q=7: total prime-order mass estimate 658.599 +- 22.737 (hit rate per sampled form 2.797e-5)
```

summary.txt

```
nonsplit q=7: samples=30000000 (random) res_shaped=839 allowed_trace_candidates=839 smooth_prime_order_hits=839 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-
sample 2000)=0/0 secs=522.0
nonsplit q=7: total prime-order mass estimate 658.599 +- 22.737 (hit rate per sampled form 2.797e-5)
```

cand_t_hist.csv (first 12 of 20 rows)

type	t	candidates
nonsplit	−75	55
nonsplit	−71	67
nonsplit	−65	47
nonsplit	−57	49
nonsplit	−45	85
nonsplit	−39	60
nonsplit	−15	52
nonsplit	−9	40
nonsplit	3	82
nonsplit	9	43
nonsplit	13	48
nonsplit	19	79

hits.csv (first 12 of 839 rows)

type	code	t	nE	smooth
nonsplit	32338162100289	25	2377	true
nonsplit	24730639447584	−9	2411	true

nonsplit	12790121929963	19	2383	true
nonsplit	2687952083866	−71	2473	true
nonsplit	7424174273030	19	2383	true
nonsplit	5216633400363	−15	2417	true
nonsplit	13591080077092	−45	2447	true
nonsplit	20627718189362	19	2383	true
nonsplit	9537333093473	19	2383	true
nonsplit	4373706492541	9	2393	true
nonsplit	8268006026927	−45	2447	true
nonsplit	27695046610450	−45	2447	true

mass_by_t.csv (first 12 of 20 rows)

type	t	nE	smooth_hits	mass_estimate	mass_stderr
nonsplit	−75	2477	55	43.1739	5.8216
nonsplit	−71	2473	67	52.5937	6.4253
nonsplit	−65	2467	47	36.8941	5.3816
nonsplit	−57	2459	49	38.4640	5.4949
nonsplit	−45	2447	85	66.7233	7.2372
nonsplit	−39	2441	60	47.0988	6.0804
nonsplit	−15	2417	52	40.8190	5.6606
nonsplit	−9	2411	40	31.3992	4.9647
nonsplit	3	2399	82	64.3684	7.1083
nonsplit	9	2393	43	33.7542	5.1475
nonsplit	13	2389	48	37.6791	5.4385
nonsplit	19	2383	79	62.0135	6.9771

Graph

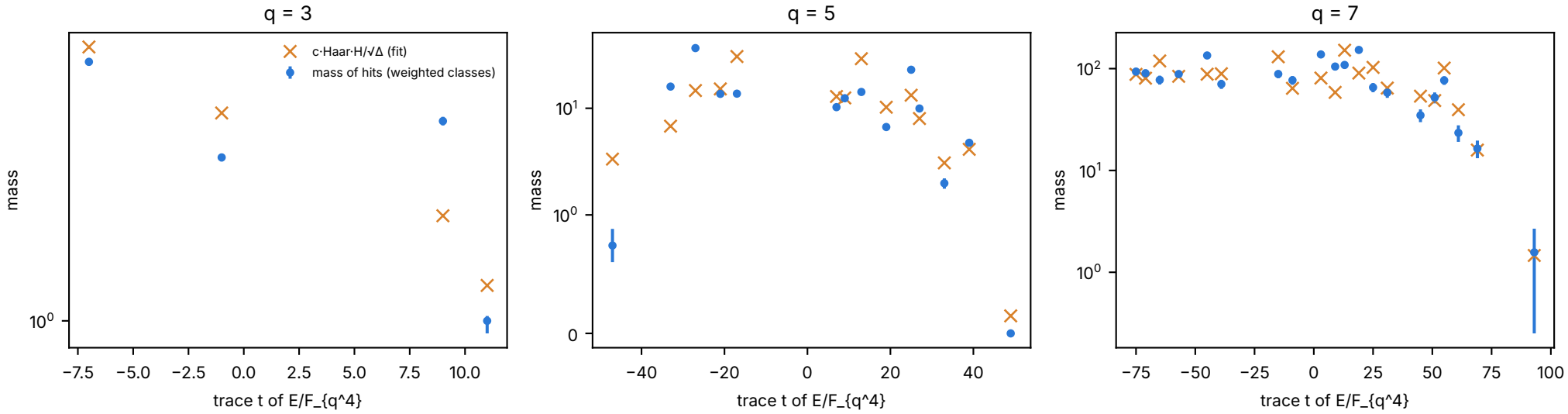


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.